

Compressed Sensing of Acoustic Cardiopulmonary Signals Using a CNN-based Reconstruction Method

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Abstract—Cardiopulmonary sounds contain a rich reservoir of vital and pathological information critical for clinical diagnosis. This paper presents a novel approach to cardiopulmonary data capturing with compressive sensing and reconstruction using a Convolutional Neural Network (CNN) based on the U-Net architecture. Applying traditional compressive sensing techniques to cardiopulmonary sounds presents several challenges. Cardiopulmonary sounds are inherently complex, with a substantial variation between captures. The traditional algorithms for compressive sensing rely on signal sparsity, whereas finding a sparse representation domain for cardiopulmonary sounds is a difficult task. Instead of finding a sparse domain manually, we propose training a convolutional encoder-decoder neural network for a pseudo-randomly undersampled set of signals without explicitly enforcing the sparsity concept. In this research, a CNN was trained for pseudo-randomly decimated input signals, evaluating a compression ratio of up to 30. The network is trained for respiratory sounds using the SPRSound dataset and for Phonocardiogram (PCG) signals using the CirCor Digiscope PCG dataset. Both these datasets have been evaluated for signal integrity after reconstruction and delivered promising results. The algorithm achieves reconstruction quality similar to that of previous research with a compression ratio three times higher than that of previous research applied to respiratory sounds. Since the principles of compressive sensing are applied in the sampling stage, the data compression requires no computation in the compression stage, and can therefore easily be implemented in low-cost edge devices.

Clinical relevance— This work enables efficient compression of cardiopulmonary sounds, maintaining high signal integrity even at three times higher compression ratios than previous methods applied to respiratory sounds. It supports low-power, portable devices for real-time monitoring, improving accessibility for telemedicine and point-of-care diagnostics in respiratory and cardiovascular conditions.

I. INTRODUCTION

Cardiopulmonary sound analysis is crucial for diagnosing and monitoring patient health, but the large data volumes generated by these recordings present significant challenges in storage, transmission, and analysis [1], [2]. This paper introduces a novel method combining Compressive Sensing

(CS) and Convolutional Neural Networks (CNNs) for efficient cardiopulmonary sound reconstruction. CS enables efficient signal capture and reconstruction from fewer samples by leveraging the signal's sparsity and the Restricted Isometry Property [3], [4]. Unlike traditional compression, which requires Nyquist-rate sampling, CS allows sub-Nyquist sampling. This work evaluates the proposed method for respiratory and heart sounds. The system overview is shown in Fig. 1, where cardiopulmonary sounds are recorded using a ModAu edge device [5] with pseudo-random sampling and sent to a computational node for CNN-based reconstruction. This study focuses on the CNN for sub-sampled sound reconstruction, with full system integration as future work. Respiratory sounds present challenges for CS due to their non-stationarity, time-varying spectral content, and irregular patterns, particularly with adventitious sounds like wheezes and crackles [6]. Additionally, noise and artifacts complicate sparse representation, potentially leading to loss of critical information during compression and reconstruction [7]. Traditional CS algorithms have shown limited success in respiratory sound reconstruction compared to other domains [8]. Heart sounds, i.e. phonocardiogram (PCG) signals, have a lower frequency range (20–150 Hz) compared to respiratory sounds (100–1500 Hz) [6], [9]. The primary challenge for PCG signals lies in noise interference and individual variability, which complicates the identification of an appropriate sparse representation for CS [10], [11]. Recent advancements in portable devices for PCG analysis [12]–[14], particularly in remote regions, have spurred interest in CS techniques that minimize energy consumption while maintaining high signal fidelity. However, these techniques must also support real-time execution on resource-constrained devices, requiring a balance between energy efficiency and diagnostic accuracy.

This study proposes combining pseudo-random undersampling with CNN-based reconstruction to address these challenges. The advantage of CS is its ability to reconstruct signals sampled at sub-Nyquist rates without computational overhead during the compression stage, making it ideal for low-cost edge devices. Fig. 1 illustrates the proposed system, where under-sampling is simulated through pseudo-random decimation. In practical applications, this can be implemented directly in the sampling process without computational overhead on devices such as the ModAu stethoscope [5]. The CNN was separately trained for respiratory sounds using the SPRSound dataset [15] and heart sounds using the CirCor Digiscope PCG dataset [16]. This allows for a direct comparison with previous methods and a detailed analysis of

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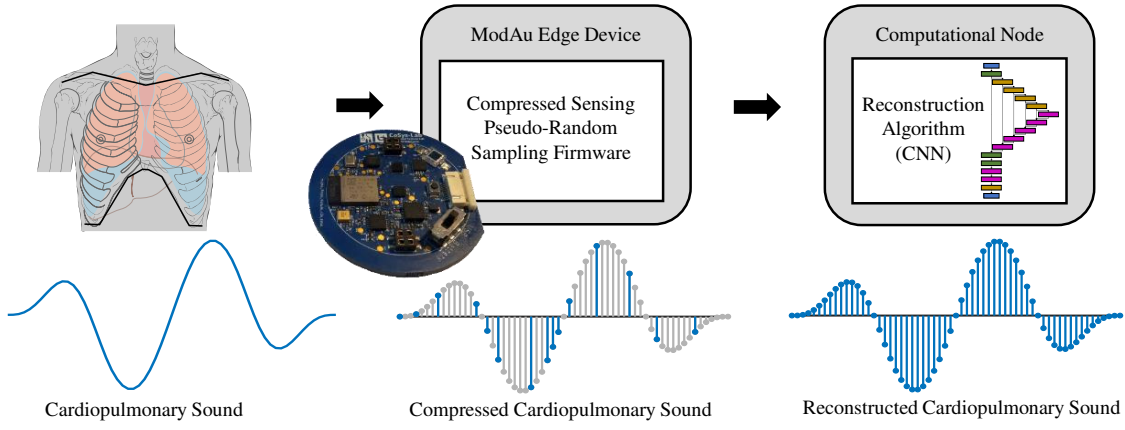


Fig. 1. Overview of the proposed system. The cardiopulmonary sounds are first recorded using the embedded ModAu edge device with pseudo-random sampling firmware for compressed sensing. Second, the compressed signals are sent to a computational node that executes the CNN reconstruction algorithm to obtain the reconstructed cardiopulmonary sounds. This paper focuses specifically on the CNN for reconstruction of the sub-sampled cardiopulmonary sound, the full integration is future work.

signal integrity post-reconstruction. Pre-processing involves compressing the data into N pseudo-random embedding vectors, as detailed in sections III-B and III-C. Given CNNs' proven success in signal processing, particularly in audio applications [17], this approach has a high potential for overcoming the limitations of traditional CS algorithms. The results of this methodology are presented in section IV, followed by discussion and conclusions in sections V and VI.

II. BACKGROUND

A. Compressive Sensing

Since the first mention of CS by Candes *et al.* [3], many CS reconstruction algorithms have been developed with the similar goal of reconstructing an estimate $\tilde{s}[k]$ of the signal $s[k]$ from a randomly sub-sampled version of this signal $s_c[k_r]$ [18]. The typical approach to reconstructing $\tilde{s}[k]$ is to solve the sparse reconstruction problem, which can be modeled as:

$$s_c[k_r] = \Phi_r \cdot \Psi_b \cdot D \cdot x + \eta \quad (1)$$

where s_c contains noisy CS measurements, Φ_r is the random sampling matrix, Ψ_b is a basis function transform from time domain into a domain where $s[k]$ is considered sparse, D is a dictionary with the basis functions in the sparse basis for $s[k]$, x is a signal with a maximum of K non-zero elements and is therefore a so-called K -sparse signal, and η is used to model Gaussian random noise. The reconstruction can then be rewritten to:

$$\tilde{s}[k] = \Psi_b \cdot D \cdot x \quad (2)$$

The existing algorithms used for estimating a reconstruction $\tilde{s}[k]$ can be divided in six approaches [19]:

- Convex Optimization (e.g., Basis Pursuit)
- Greedy Iteration (e.g., Orthogonal Matching Pursuit (OMP))
- Thresholding (e.g., Iterative Hard Thresholding)
- Combinatorial (e.g., Heavy Hitters on Steroids)

- Non-Convex Optimization (e.g., Iterative re-weighted Least Squares)
- Bayesian (e.g., Bayesian Compressive Sensing)

Each of these approaches has its own set of limitations, such as complexity and inaccuracy. The development of a universal method that can be adapted to every application while maintaining a certain quality of reconstruction is therefore desirable.

B. Traditional Approaches to Cardiopulmonary Sound Compression

The results achieved by the algorithms mentioned in section II-A in other domains, differ significantly from the results currently achieved on respiratory sounds. As explained in the work of Kahya [7], respiratory sounds are significantly more challenging to represent sparsely than other physiological signals due to time-varying spectral content and irregular patterns. For this reason, the problem lies in finding a suitable dictionary for Eq. 2. To compare and evaluate our proposed method, two traditional methods are also applied on the use-case of respiratory sounds. First, the work of Chen *et al.* [8] uses a Bernoulli random matrix to determine the samples to be used. These samples are transformed using the Optimized Discrete Cosine Transform (ODCT). An OMP algorithm is then used as an iterative reconstruction algorithm. Second, the work of Wohlberg *et al.* [20] describes a convex optimization algorithm for the recovery of under-sampled signals. There are no previous applications of this method on respiratory sounds, but due to promising results in other domains [21], this method was also selected for comparison.

To the best of our knowledge, the CS reconstruction of PCG signals using traditional methods has not been thoroughly investigated in the literature. Due to the limited performance observed for respiratory sounds, we opted not to evaluate convex optimization methods. As such, we only present the statistical metrics with the uncompressed signal as a reference, as well as a clinical validation using a well-established

segmentation algorithm to evaluate our proposed method on PCG signals.

III. MATERIALS AND METHODS

A. Respiratory and Heart Sounds Datasets

1) *SPRSound Dataset*: The open-source pediatric respiratory sound database (SPRSound) [15] consists of 2,683 records and 9,089 respiratory sound events from 292 pediatric participants, dedicated to respiratory analysis of the pediatric population. The sound fragments have a maximum length of 17.5 seconds and a sample rate of 8 kHz.

2) *CirCor Digiscope PCG Dataset*: The CirCor Digiscope PCG dataset [16] contains a total number of 5,272 heart sound recordings collected from the main four auscultation locations of 1,568 subjects, aged between 0 and 21 years (mean $\pm$ STD = 6.1 ± 4.3 years), with a duration between 4.8 to 80.4 seconds (mean $\pm$ STD = 22.9 ± 7.4 s). The total recording time is more than 33.5 hours of recording at a sampling rate of 4 kHz.

B. Data Compression

The first step in preparing the signals for the CNN is to compress the data using CS. In literature, CS is often implemented using pseudo-random compression of previously sensed data obtained by Nyquist-conform sampling [3], [4], [8], [22]. In these cases, a pseudo-random set of samples is used from the entire captured dataset to maximize incoherence between the signal and its sampling. However, solid results have also been achieved with an embedded implementation of CS using a moving pseudo-random sampling mechanism [23], [24]. Since acquisition devices are generally embedded devices with memory and processing constraints, a moving pseudo-random sampling method is preferred to select the random sample times for the compressed signal. A further benefit of this approach is that the compression is simply a pseudo-random a priori decimation, which poses a clear advantage compared to the overhead of traditional compression methods. To discover the limits of our method, pseudo-random decimation is used to simulate the moving pseudo-random sampling (with factors between 2 and 30). This factor will further be referred to as the compression ratio CR . For the calculation of the pseudo-random amount of skipped samples d , presented in Eq. 3, the assumption is made that for an under-sampling factor CR , the dataset is still sufficiently random if d is uniformly distributed between 0 and $(2 \cdot CR - 1)$ skipped samples. As such, d is the remainder of a modulo division, where the dividend x is a 16-bit pseudo-random number that changes for each calculation of d and the divisor equals $2 \cdot CR$ [23], [24].

$$d \equiv x \mod (2 \cdot CR), \quad x \in \{0, 1, \dots, 2^{16} - 1\} \quad (3)$$

C. Pre-processing for the Reconstruction CNN

Practically, the original (non-compressed) signals $s[k]$ in the SPRSound dataset have a maximum size of 139,904 samples. Fragments shorter than 139,904 were zero-padded. The

CirCor dataset contains 59 files containing more than 139,904 samples. For simplicity, these were cropped to 139,904 samples. The rest of the signals were zero-padded, similar to the SPRSound dataset processing. After reconstruction, the signal is restored to its original length by removing the zero padding. To improve the learning capacity of the CNN, embedding vectors are used. In practice, these are different under-sampled versions of the same signal. Because we are taking multiple independent pseudo-random embedding vectors from the same signal, certain samples from the original signal may be sampled multiple times by separate embeddings. Therefore, the actual total compression ratios can be even higher than the mentioned total compression ratios in this paper. To simulate the compressed sensing in the embedded device, N pseudo-randomly sub-sampled embedding signals $s_c^n[k_r]$ were sampled from $s[k]$. These sub-sampled signals have a compression ratio CR_s that is calculated from the total compression ratio CR_{tot} and the number of embedding signals N using this formula:

$$CR_s = CR_{tot} \cdot N \quad (4)$$

Before the embedding signals $s_c^n[k_r]$ are sent to the CNN for reconstruction, they are first linearly interpolated to $s_b^n[k_i]$ so that they are sampled regularly and have a fixed length of 34,976 samples. This interpolation operation introduces artifacts in the frequency spectra of S_b^n , which are resolved by the CNN. We chose not to interpolate to the original 139,904 samples, to reduce the input size of the CNN by factor 4. The resulting input dimensions of the CNN are therefore 34,976 by N . Further evaluation of the optimal network size is future work.

D. Convolutional Neural Network for Signal Reconstruction

To circumvent the learning step of a dictionary from Eq. 1 and the iterative process used in traditional methods, we apply a deep learning model to reconstruct the signal. Mathematically, we apply a neural operator G to find the reconstruction $\tilde{s}[k]$ from these N interpolated embedding vectors $s_b^n[k_i]$.

$$\tilde{s}[k] = G(s_b^1[k_i], \dots, s_b^N[k_i]) \quad (5)$$

Recent advancements in deep learning have spurred innovative approaches to address the challenges traditional compressive sensing methods encounter. The technique proposed in this paper involves training an encoder-decoder network based on convolutions with skip connections. This network is inspired by the U-Net architecture [25] and aims to reconstruct the under-sampled signals. Encoder-decoder networks have demonstrated exceptional capabilities in learning complex mappings between under-sampled measurements and full-dimensional signals [26]. Fig. 2 presents an overview of the proposed reconstruction network. The CNN is constructed with 15 blocks of three different types:

- 1) a regular convolutional block that contains a 2D convolutional layer;

- 2) a convolutional/pooling block that consists of an average pooling layer followed by a 2D convolutional layer;
- 3) and a transposed convolutional block composed of a transposed 2D convolutional layer.

Next to the layers mentioned above, each block starts with a batch normalization layer (except for the first block) and ends with a Leaky ReLU activation function (except for the last block). The last three blocks were implemented to reverse the decimation of factor 4 in the interpolation step of the pre-processing stage. Therefore, the output dimension equals the original (padded) signal length. The activations of the skip connection are reintroduced in the network by concatenating the activations in the channel dimension. To train and validate our algorithm, we trained the CNN separately for the two datasets considered. To train the SPRSound dataset the data was split into 80% train and 20% test data. Whereas the CirCor PCG data was split into 60% train and 40% test data, to ensure a sufficient amount of data to properly train the segmentation CNN as well. Both training datasets are augmented to a total of ten times the original files using white noise and by shifting the signals in time. 3% of the training data is used for validation. The networks are trained using the Adam optimizer with an initial learning rate of 0.5×10^{-4} , a learning rate drop factor of 0.97, a gradient decay factor β_1 of 0.9, and a squared gradient decay factor β_2 of 0.999. The network used for $N = 2$ embedding vectors had approximately 3 million learnable parameters. The reconstruction of the whole SPRSound dataset, comprising approximately 180 minutes of respiratory sounds, took circa 110 seconds on a computer with a single NVIDIA RTX 2060 GPU running Matlab R2024a. Reconstructing the CirCor Digiscope dataset took instead circa 169 seconds. The code used to train and validate this network can be publicly accessed at <https://cosysgit.uantwerpen.be/TStas/cs-rs-cnn>.

E. Evaluation metrics

1) *Respiratory sounds*: The proposed CS-based compression and CNN-based reconstruction algorithms are evaluated using the SPRSound respiratory dataset and the following evaluation performance metrics: Compression Ratio (CR), Relative Root Mean Square Error ($RRMSE$), and Correlation Coefficient (CC). These metrics are defined as:

$$CR = \frac{\text{Size of initial signal}}{\text{Size of compressed signal}} \quad (6)$$

$$RRMSE = \sqrt{\frac{\sum_k (\tilde{s}[k] - s[k])^2}{\sum_k s[k]^2}} \quad (7)$$

$$CC = \frac{\text{Cov}(s[k], \tilde{s}[k])}{\sigma(s[k])\sigma(\tilde{s}[k])} \quad (8)$$

2) *Phonocardiogram Signals*: To evaluate the reconstruction quality of PCG signals, the same metrics are used as for respiratory sounds. The reconstructed heart sound signals are

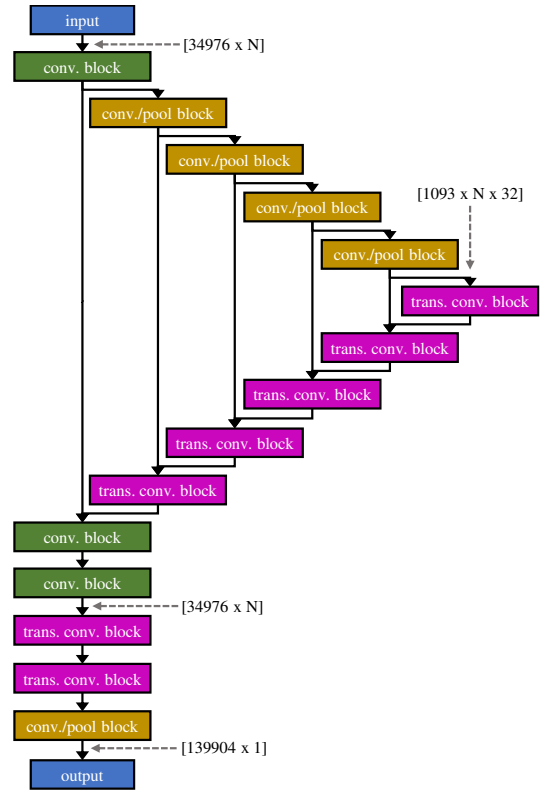


Fig. 2. Overview of the proposed reconstruction CNN. The regular convolutional, convolutional/pooling, and transposed convolutional blocks have respective green, yellow, and purple colors. The number of activations is noted in square brackets with N denoting the number of embedding vectors.

then segmented using an existing and well-established CNN-based heart sound segmentation algorithm [12], [13], [27]. The segmentation is evaluated for both original and reconstructed data using the following metrics: Recording Accuracy (A_R), defined as the fraction of samples in the output sequence $\tilde{s}(t)$ that are correctly allocated to the corresponding state label in the ground truth sequence $s(t)$, Positive Predictive Value (P_+) in Eq. 9 and Sensitivity (S) in Eq. 10:

$$P_+ = \frac{T_p}{T_p + F_p} \quad (9)$$

$$S = \frac{T_p}{T_{tot}} \quad (10)$$

In Eq. 9 and 10 a true positive (T_p) is counted when the center of an S1 (or S2) sound in the estimated sequence is $\tilde{s}(t)$ is closer than 60ms from the corresponding sound in the ground-truth sequence $s(t)$. All others are considered false positives (F_p), and T_{tot} is defined as the total number of S1 and S2 sounds in the ground-truth sequence $s(t)$.

IV. RESULTS

A. Respiratory Sounds

The following input scenarios were evaluated to determine the achievable limits. First, the model was trained for three epochs with a fixed compression ratio of five for one to five

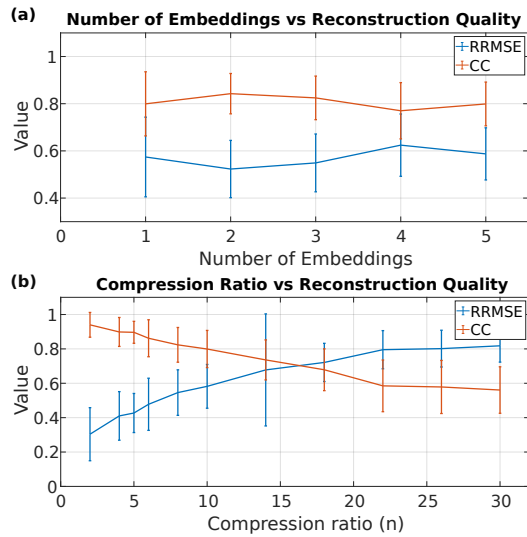


Fig. 3. Test results for model parameter variations after training each model for three epochs. The reconstruction quality is presented using the *RRMSE* and the mean *CC* between the original signal and its reconstructed version. The line graph connects the mean calculations from the reconstructed dataset, where the vertical range represents the standard deviation. (a) Evaluation of the optimal number of embedding vectors N to use as CNN input. The use of two embedding vectors is chosen as optimal and used in further evaluations from this evaluation. (b) Evaluation of the compression ratio in function of reconstruction quality.

embedding vectors. Following these results, the compression ratio was varied between 2 and 30 using a fixed number of two embedding vectors. The results can be seen in Fig. 3. The presented evaluation metrics are the result of applying our method to the SPRSound test dataset. To test the limitations of our method, a compression ratio of 30 is used to achieve a mean *CC* of 0.60 and a mean *RRMSE* of 0.79. To compare the achieved results with the state-of-the-art, two alternative methods were used for comparison. The first is Convolutional Basis Pursuit Reconstruction using Dictionary Learning (CBPRDL) [28]. This method was chosen because solid results have been achieved with this approach [21]. The second method for comparison is a combination of ODC and OMP. [8]. Our method, combining Compressive Sensing and a Convolutional Neural Network (CS-CNN), achieves a similar reconstruction quality as ODC-OMP [8] for a compression ratio three times higher. A comparison between the original signal and its reconstruction using a compression ratio of 10 is presented in Fig. 4. As the comparison metrics in Table I have already indicated, the reconstructed signal is a close match to the original signal. Whereas the amplitude is slightly mismatched in the highest peaks, the trend of the original signal is properly maintained. The spectrogram results indicate that the frequency content is very reliably reconstructed.

B. Phonocardiogram Signals

Observing Table I, it is evident that as the CR increases, the CC decreases while the RRMSE increases. This is likely due to signal peaks caused by external noise that are not reconstructed but would typically be removed during pre-processing

before segmentation. Additionally, the spectrograms from the reconstructed PCGs reveal that much of the high-frequency information is lost, which is reflected in the values reported in Table I. Nevertheless, information below 150 Hz appears to remain largely unaffected.

To apply the same algorithm to heart sounds, we used $N = 2$ number of embeddings for the reconstruction CNN. The segmentation evaluation is done in two ways. First, the segmenter [13] is used in its original trained state, configured with $n_{enc} = 4$ encoder/decoder layers, $N = 64$ input window, and $n_0 = 4$ base number filters. While in [13] the authors identify $n_0 = 8$ as the configuration yielding optimal performance, only the pre-trained weights for $n_0 = 4$ were available for this work. However, according to the findings and statements in the cited paper, the reduction in performance associated with this configuration is negligible. The use of the model trained exclusively on original PCG data serves as an indication of the actual conservation of signal properties in an absolute sense. The results from this analysis are presented in Table II-A and are based on the 40% of data already used for testing the reconstruction CNN. The reported values are from a single-pass inference, and thus no standard deviation is provided. Second, the segmenter is retrained employing the reconstructed PCG signals from the 40% of data, used for evaluating the reconstruction CNN, and a 10-fold cross-validation as in [13]. The results from this evaluation are presented in Table II-B. As can be seen, the model's performance has been successfully restored, even at a CR of 30. Notably, both at CR 10 and 30, the results are still comparable to those achieved with original PCG input. The filtering effect of high-frequency information may

TABLE I
RECONSTRUCTION EVALUATION

Method	CR	CC $\uparrow$	RRMSE $\downarrow$
RESPIRATORY SOUNDS			
CBPRDL [28]	3	0.17	0.98
ODCT-OMP [8]	3.5	0.86	0.50
CS-CNN	4	0.90 $\pm$ 0.08	0.41 $\pm$ 0.14
CS-CNN	10	0.85 $\pm$ 0.08	0.50 $\pm$ 0.12
CS-CNN	30	0.60 $\pm$ 0.13	0.79 $\pm$ 0.10
PHONOCARDIOGRAM			
CS-CNN	4	0.64 $\pm$ 0.17	0.74 $\pm$ 0.15
CS-CNN	10	0.46 $\pm$ 0.18	0.86 $\pm$ 0.11
CS-CNN	30	0.32 $\pm$ 0.14	0.94 $\pm$ 0.06

TABLE II
PCG SEGMENTATION EVALUATION [13].

Method	CR	A_R (%)	P_+ (%)	S (%)
A – SEGMENTATION CNN NOT RETRAINED				
Original PCG	-	92.7	97.6	97.3
CS-CNN	4	91.6	96.9	96.2
CS-CNN	10	89.0	95.2	93.3
CS-CNN	30	83.0	90.9	86.4
B – RETRAINED SEGMENTATION CNN				
Original PCG	-	92.1 $\pm$ 1.4	96.2 $\pm$ 1.3	95.7 $\pm$ 1.5
CS-CNN	4	92.1 $\pm$ 1.3	96.4 $\pm$ 1.3	95.8 $\pm$ 1.3
CS-CNN	10	91.6 $\pm$ 1.0	96.1 $\pm$ 1.0	95.1 $\pm$ 1.1
CS-CNN	30	90.8 $\pm$ 1.5	96.0 $\pm$ 1.1	94.7 $\pm$ 1.4

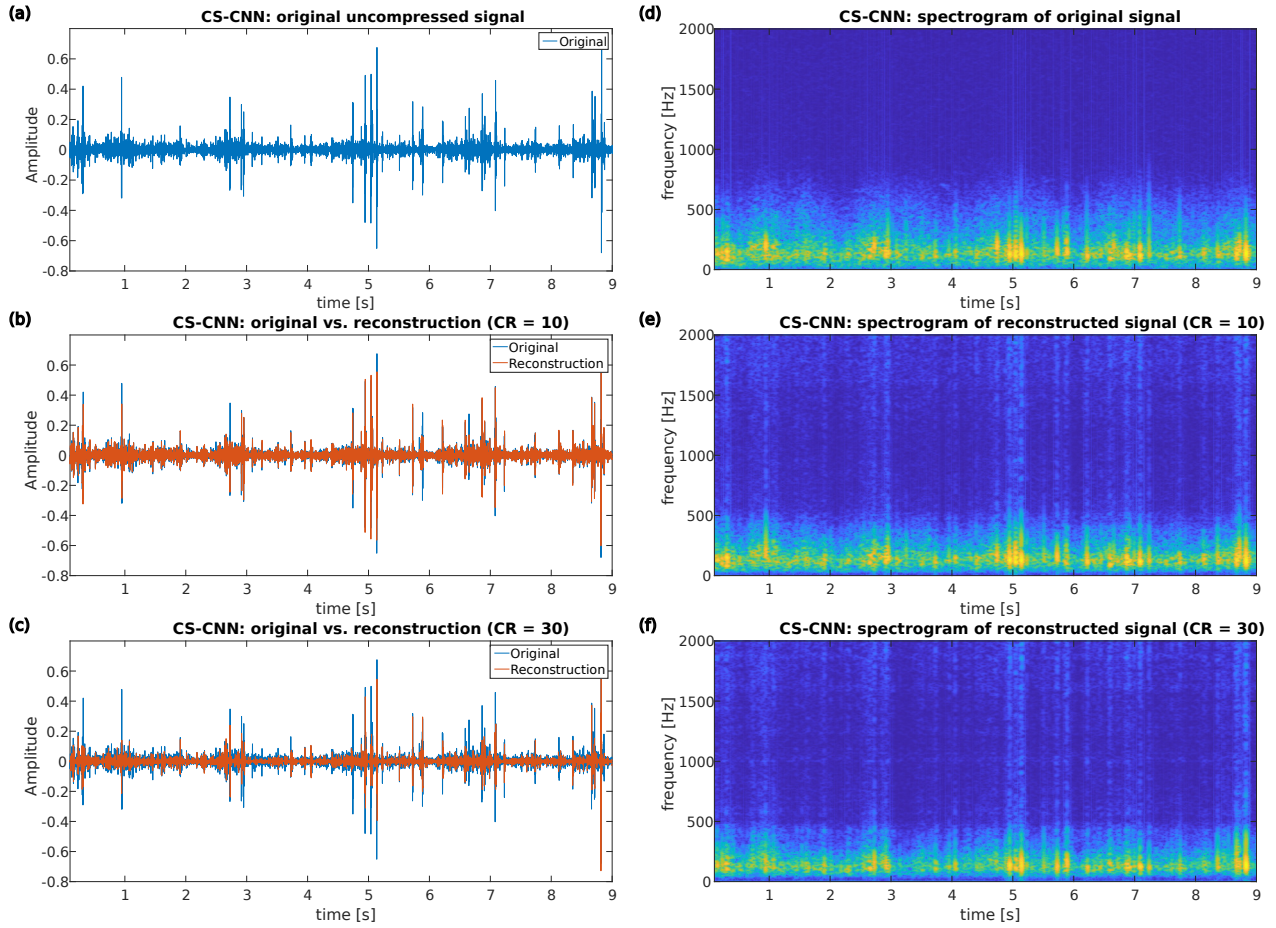


Fig. 4. Comparison between original data and its reconstruction using the CS-CNN approach on the SPRSound dataset [15] using a compression ratio of 10 and 30. (a) The original signal in the time domain. (b) The original signal in comparison with the reconstruction using CR = 10 in the time domain. (c) The original signal in comparison with the reconstruction using CR = 30 in the time domain. (d) Presents the spectrogram within the region of interest for respiratory sound analysis of the original signal. (e) Presents the spectrogram of the reconstructed signal using a compression ratio of 10. (f) Presents the spectrogram of the reconstructed signal using a compression ratio of 30.

have contributed to enhancing the robustness of the four features provided as input to the segmenter, making them less sensitive to external noise. Consequently, after retraining, the model appears to have learned to distinguish heart sounds with greater stability, as evidenced by the minimal decline in performance metrics for all the CRs.

V. DISCUSSION

The proposed method overcomes limitations of traditional CS methods by relaxing strict sparsity constraints, making it more applicable to real-world scenarios. However, the computational demands of the CNN architecture could limit its feasibility for portable medical systems due to the large number of parameters required. While hardware devices such as FPGAs, microDSPs, and smart sensors can accelerate signal processing, optimizing techniques to reduce computational demands remains crucial for improving real-time performance and power consumption in resource-constrained applications.

This complexity highlights the need for parameter-efficient models, such as lightweight CNNs or neural architecture search techniques, to balance performance and resource constraints. The trade-off between computational cost and

efficiency is notable. Unlike traditional CS methods that rely on iterative optimization, CNNs shift computational effort to training, allowing faster inference but requiring significant resources overall. Comparing the computational efficiency of CNN-based and traditional CS methods can guide decisions for deployment.

Alternative architectures, such as Recurrent Neural Networks (RNNs) or transformer-based models, could better capture temporal dependencies in cardiopulmonary signals, while hybrid models combining CNNs with other networks may improve performance without excessive resource demands. The reconstructed PCGs showed a filter-like effect, likely caused by frequency-dependent weighting during reconstruction. This may suppress key features, affecting diagnostic accuracy, particularly for high-frequency signal characteristics. Further investigation and applying domain-specific constraints to preserve fidelity across all frequencies could address this concern.

VI. CONCLUSION AND FUTURE WORK

This paper introduces a pseudo-random sub-sampling compression technique combined with CNN-based reconstruction,

offering significant improvements over traditional methods. The proposed CS-based compression can be implemented without computational overhead during data acquisition at the stethoscope level, while reducing power consumption as shown by [8]. The CNN effectively addresses the challenge of sparsely representing cardiopulmonary data, with results demonstrating up to three times higher compression efficiency than previous methods when applied to respiratory sounds [8]. While a trade-off between compression ratio and reconstruction quality exists, the practical benefits of the method are clear. By implementing the technique in portable, low-power devices, this work enables efficient real-time monitoring with reduced processing demands. The CS approach, particularly in embedded devices like the ModAu [5], reduces power consumption and data transfer requirements while maintaining signal quality. Future work will focus on embedding the proposed sampling method into digital auscultation devices, further enhancing the practical usability of both the device and the research findings.

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